

L06 — Chatbot II

Prompting, RAG, and chatbot evaluation

Study summary generated from lecture slides. ETH AI for Digital Characters.

Prompting

- Roles: system (persona), user (input), assistant (history)
- In-context learning: examples at inference, no gradient updates
- Structured outputs: JSON schema, function calling / tool use
- ReAct: Thought -> Action -> Observation loop
- DSPy: automatic few-shot example selection

Retrieval Augmented Generation (RAG)

- Solves: hallucination, stale/private knowledge
- Pipeline: chunk documents -> embed -> retrieve -> augment prompt -> generate
- Chunking: fixed-size, recursive, semantic
- Sparse retrieval: tf-idf, BM25 + inverted index
- Dense retrieval: bi-encoder (SBERT), ColBERT (token MaxSim)
- Vector DBs: FAISS, Pinecone, Milvus, Weaviate (ANN search)
- Reranking: bi-encoder (fast, top-100) -> cross-encoder (precise, top-5)
- Evaluation: grounding score, answer relevance

Chatbot evaluation

- Qualitative: MOS (1-5 rating), A/B preference tests
- Quantitative: perplexity, BLEU, cosine similarity
- Perplexity: $P = 2^{-(1/N * \sum \log_2 P(w_i))}$ - lower is better
- LLM-as-a-Judge: strong LLM rates responses
- Benchmarks: Chatbot Arena (Elo), MMLU, Stanford HELM